C964: Computer Science Capstone Task 2

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Part A: Letter of Transmittal

Letter of Transmittal

Subject: Heart Failure Detector

April 30th, 2024

Dr. John Doe, CTO

Lenox Hill Heart Hospital

130 E 77th St, New York, NY 10075

Dear Dr. John Doe,

As seen at your Heart Hospital, with a growing aging population worldwide and an increase in contributing poor health habits, cardiovascular health has become more crucial than ever. At the same time, many medical teams around the country are understaffed and lack the necessary medical tools, technology, and equipment. This results in an array of issues including longer patient waiting times, inaccurate diagnoses, and medical errors. As cardiovascular health services grow in demand, the medical practitioners at your hospital seek better technology to assist them in making accurate and well-informed decisions to better support your patients.

To resolve this problem, I propose the development of an application utilizing machine learning to predict if a patient is likely to suffer heart failure using the patient's medical data.

This application would use machine learning to train on medical data and be able to make a prediction based on the data inputted in a form. A medical practitioner would be able to access this application using the internet on their device, fill out a form with relevant fields such as the patient's age, gender, and blood test levels, and then instantly view the result of whether they are likely to have a heart failure or not. This product would benefit your hospital by providing a tool that will automate evaluating a patient for a risk, aiding in the overall treatment efficiency and accuracy.

This proposed project would be developed over just seven weeks using a waterfall methodology with clearly defined goals and a streamlined process to help keep it on track. It would have an up-front funding requirement of $43,700 to cover the costs for contract staff, hardware needed, and hosting the application on the internet. The development tools themselves such as programming languages would be at no cost since many are free and publicly available, decreasing the overall funding costs. It also would use open-source data with a few hundred samples that do not contain any ethical or legal concerns, making the development easier to facilitate.

Additionally, as a Software Engineer, I have worked with a state medical organization for over 3 years, allowing me to gain insight into medical practitioners' needs. These experiences would allow me to make the best technological decisions while keeping in consideration your medical staff's practices and preferences.

Given the project's potential to help your medical staff and patients as well as streamlined development in a short duration of time, we invite you to consider our project proposal. Let us know if you have any further questions or concerns. We look forward to hearing from you soon!

Sincerely,

Ruslan Abramov

Software Engineer

Project Recommendation

Summary of the Problem

As cardiovascular health services grow in demand, many medical teams around the country are understaffed and need tools to help them automate and improve the efficiency of medical processes such as predicting the risk of heart failure. Over the past decades, the population has grown and faced an increase in contributing poor health habits, posing a demand for heart health services. At the same time, many hospitals around the country are severely understaffed and lack the necessary medical tools, technology, and equipment. This results in an array of issues including longer patient waiting times as well as inaccurate diagnoses and medical errors. Such issues further demonstrate that medical practitioners need better technology to assist them in making accurate and well-informed decisions, supporting the patient as well.

Data Product Benefits

This product would benefit the medical practitioner team by providing a tool that will automate evaluating a patient for a risk, aiding in the overall treatment efficiency and accuracy. An online form with an instant result would be faster to fill out than taking the time to manually consider each of the patient's factors, resulting in faster treatment of the patient and alleviating the medical practitioner's workload. Additionally, this application would use a machine learning model with a projected accuracy rate of at least 75%, which would also help increase accuracy and diagnosing consistency.

Outline of The Data Product

To resolve this problem mentioned earlier, I propose a data product solution of an application utilizing machine learning to predict if a patient is likely to suffer heart failure using the patient's medical data. This application would be available online, use machine learning to train on sample data and make predictions, and contain a heart failure prediction form. This form would include fields for patient data such as their age, gender, blood test results, a button to submit it, and a place to view the submission results stating whether the patient is predicted to have heart failure or not. A medical practitioner would be able to access this application using the internet on their device, fill out a form with relevant fields, and then instantly view the result, and take the appropriate actions to treat the patient accordingly based on their medical protocol.

Description of The Data Used

The data that would be used will be a sampled dataset containing raw medical data from an online free source, Kaggle. The data will be downloaded directly from the website as a .csv spreadsheet with multiple columns for various medical factors and one column indicating whether that person had heart failure. All fields would be represented in numerical form, with a row for each patient. This data would be downloaded one time and stored in the same folder as the main Python application file.

Objectives and Hypotheses

The main objective of this project is to provide better accuracy and efficiency of the heart failure detection process by building a publicly available usable application with a form that uses machine learning to process form inputs of medical data and output the heart failure prediction. The hypothesis is that if a heart failure detection model is implemented using a machine learning logistic regression model, it will have an average accuracy score of at least 75% when presented with sample cases.

Outline of The Project Methodology

The industry-standard methodology that will be used is the waterfall methodology because it would set concrete goals and a timeline that would be easier to manage, ensuring successful project development and completion consistent with the original goals.

The waterfall development steps would include the following:

1) Gathering of requirements related to the medical practitioners' needs and outlining some of the technical constraints such as software versions that need to be used
2) Design and planning of the application such as the type of algorithm that would be used
3) Implementation to train the machine learning model and write additional code in Python
4) Verification to test the application usability including the machine learning model accuracy and form functionality
5) Deployment of the application and maintenance to fix any issues that the first set of users may face in the future

Funding Requirements

The total sum for this project will be $43,700, which would cover three contract full-time staff salaries, necessary equipment, and hosting of the application. The development tools such as integrated development environment and programming language would be free since they are available for free on the internet. The data would be free as well since it is also publicly available for free. The project will take place over seven weeks, with three team members working full-time five days a week. The annual maintenance costs after the deployment will be $1,200-$5,200 per year.

Impact of The Solution on Stakeholders

The stakeholders, including medical practitioners, patients, and the hospital, would benefit from the application allowing better efficiency. The medical practitioners would benefit as it would be easier and faster to process patient data in order to make a prediction regarding their health, which would allow practitioners to focus more on the treatment aspect and have a lighter workload. Additionally, since this application can help save time, the patients would also benefit by having shorter waiting times. As a whole, having more patients be processed in a fast and easy manner can help increase the hospital's performance and reviews, which in return can increase its profits and funding.

Ethical and Legal Considerations

This dataset will not have any ethical or legal concerns because it will be obtained from a free public source and will include anonymous patient data. It does not contain any personally identifiable or sensitive information such as names, birthdays, SSNs, locations, or patient IDs. Additionally, the personal information that is present, such as age and gender, is not sufficient to identify a particular person.

Expertise Relevant To The Solution

My expertise is relevant to this project as I have both software engineering experience and background healthcare knowledge. I have three years of experience as a Web Developer and have taken university courses in software and AI development. Additionally, as a Web Developer, I have worked with a state medical organization, so I have gained more insight into medical practitioners' needs. These experiences would allow me to make the best technological decisions while keeping in consideration medical staff's practices and preferences.

Part B: Project Proposal Plan

Problem

The problem that will be addressed is that as cardiovascular health services grow in demand, medical practitioners need better technology to assist them in making accurate and well-informed decisions and improve efficiency regarding heart failure detection. With a growing aging population and an increase in contributing poor health habits, cardiovascular health services are increasingly in sought-after. However, at the same time, many hospitals are severely understaffed and lack the necessary medical tools, technology, and equipment. This results in an array of issues including longer patient waiting times as well as inaccurate diagnoses and medical errors, placing a need for tools that help automate and improve medical care, resulting in better efficiency.

Customers and Product Benefits

The customers in this project will be the medical staff at a hospital specializing in cardiovascular health (heart issues). These medical providers would be seeking technological tools that can allow them to automate some of the time-consuming tasks such as analyzing patient data to predict their health outcome.

This application will benefit the client because it will provide a tool that will automate evaluating a patient for a risk, aiding in the overall treatment efficiency and accuracy. An online form with an instant result will be faster to fill out than taking the time to manually consider each of the patient's factors, resulting in quicker processing times. Additionally, this application uses machine learning with a high accuracy rate, which would also increase accuracy and diagnosing consistency as well.

Existing Gaps in The Data Products

This product will not aim to replace or modify any of the current tools because its purpose is to automate tasks that are done in person without a particular system or tool. The way medical practitioners detect a risk of heart failure varies greatly based on the specific hospital or clinic practice and what type of issues the patient is already coming in with, so there is not one specific method for doing so.

Data Source and Processing

The data that will be used is a spreadsheet with columns for multiple medical factors and a column if the person had heart failure, all represented in numerical form, with a row for each patient with at least a few hundred samples. The raw data will be collected from an online free source that contains raw medical data, Kaggle.

The data will be collected, processed, and managed in the following stages:

1) Requirements: the project will lay out data requirements and constraints such as containing numerical data only and having multiple columns for independent variables and only one column for a dependent variable
2) Before implementation: the data will be downloaded from the source as a .csv file and checked to ensure that all of its fields are filled out and do not contain any missing data
3) Implementation: the data will be loaded into the application file as a data frame and split into arrays for independent and dependent variables using Python
4) Deployment: the data will be deployed and stored in the same folder as the main application file
5) Maintenance: since the application deployment, the data is expected to stay the same but will be checked periodically to ensure that the data file has not become unexpectedly corrupted or deleted

Data Benefits

The data will meet the needs of this project because it will provide multiple independent variables of medical factors affecting the dependent variable of whether the person had heart failure. Having a dependent variable fall into one of two categories makes it easy to train a machine-learning model with logistic regression since logistic regression easily allows the prediction of one of the two results.

This dataset will include columns for medical factors such as age, smoking, presence of medical conditions such as diabetes, and various blood test level results such as serum creatinine. This wide variety of independent variables will aid in the accuracy of the model as it can use more factors to rely on for prediction. Additionally, this dataset will contain at least a few hundred rows of patient data, which will make the model more accurate by providing it with more data to train on.

Ethical Concerns

This dataset will not have any ethical or legal concerns because it will include anonymous patient data and will not contain any personal information such as names, birthdays, or patient IDs. Additionally, the personal information that will be present such as age and gender is not sufficient to identify a particular person.

Methodology

The industry-standard methodology that will be used is waterfall methodology because it would set concrete goals and timelines that would be easy to manage, ensuring successful project development and completion consistent with the original goals. These steps would include gathering requirements, designing, implementation, verification, deployment, and maintenance. The detailed tasks for each of these steps are included in the Implementation Plan section.

Deliverables

The deliverables will include a finished application with a form for medical information that when submitted, will state if a patient will have a heart failure based on the machine learning model's prediction and contain multiple data visualization tools. This application should be accessible online using a link and contain the necessary interface for the form, such as the form fields, buttons, and output for the user to use. This final product will then be shared with the client using a link and should be delivered meeting its final deadline.

Implementation Plan

The steps for this project are as follows:

1. Gathering requirements:
    a. Meet with stakeholders (the hospital's leading medical staff) to ensure that the project aligns with their expectations.
    b. Define technical constraints such as software versions that need to be used and how the application should be accessed.
    c. Gather data requirements such as which fields should the dataset contain and where it will be sourced from.
2. Design and planning:
    a. Plan which type of algorithm will be used for machine learning, such as linear or logistic regression.
    b. Plan which specific data should be used and how and where it will be stored.
    c. Make a software design including the basic functions of the application that will meet the requirements.

3. Implementation:
    a. Process the data to ensure it does not have any formatting or spelling variations or missing fields, fix it if needed, and load it into the application directory.
    b. Develop a machine-learning model by training it on the data provided using logistic regression.
    c. Develop a form that contains user input of medical data and output of the prediction.
    d. Create data visualization tools such as charts for descriptive methods to help the user visualize the data trends.
4. Verification:
    a. Test the model by feeding it test data and comparing its prediction results to the actual results to ensure that it meets accuracy goals.
    b. Test the form's usability by entering different types of inputs and checking that it outputs the appropriate response.
    c. Test the application as a whole to ensure that no parts of the code result in warnings or errors.
5. Deployment and maintenance:
    a. Deploy the application onto the hosting system and share it with customers.
    b. Continue maintenance throughout the software lifecycle, releasing bug fixes and software upgrades as needed.

Evaluation and Verification Plan

The following is a list of verification methods used at each development stage:

1. Gathering requirements: ensure that the project requirements and goals of the development team align with stakeholders' (the hospital's medical practitioner staff) needs.
2. Design and planning: verify that the software design and data that will be used are compatible with the project's requirements and meet technical constraints.
3. Implementation:
    a. Ensure that the data does not have any formatting or spelling variations or missing fields.
    b. Check that the form and data visualization tools are functional and visible to the user.

The following is a list of verification methods upon completion of the project before the deployment (verification stage):

1. Test the machine learning model by feeding it test data and comparing its prediction results to the actual results to ensure that it meets the average accuracy goal of at least 75%.
2. Test the form's usability by entering different types of inputs and checking that it outputs the appropriate response.
3. Test the application as a whole to ensure that no parts of the code result in warnings or errors.

The following is a list of verification methods after the project after the deployment (deployment and maintenance stage):

1. Ensure that the users can view and access the application from the internet.
2. If any bugs or issues occur, use the maintenance budget and resources to update the application to fix it.

Resources and Costs

The resources and costs are included in the table below. The total sum for this project will be $43,700 including contract full-time staff salaries, equipment, and hosting of the application. The maintenance after deployment will be $1,200-$5,200.

Resource	**Cost**	**Explanation**
AI Software Engineer: specializes in AI and machine learning and would be able to create an ML model	$15,400	$55/hour * 40 hours / week * 7 weeks
Full-Stack Software Engineer: assists AI Software Engineer in other development tasks such as testing and deploying	$14,000	$50/hour * 40 hours / week * 7 weeks
Project Manager: ensures that the project stays on track, does planning, and communicates with other groups regarding decisions such as project approval	$11,200	$40/hour * 40 hours / week * 7 weeks
Hardware: personal computers for developing the application and communicating in a team	$3,000	$1,000/laptop * 3 people
Software: ● Programming language: Python ● Libraries: Pandas, Sklearn, Matplot, Ipywidgets ● IDE: Visual Studio Code	$0	Those tools are free and publicly available.
Hosting: allows the application (includes both the main Python file and the data file) to be publicly available on Amazon Web Services	$100	Monthly hosting fee
Data: dataset used for machine learning	$0	The data can be downloaded for free.
Internet connection and electricity: for working on the project	$0	The team members will work remotely at their own expense.
Total Upfront Costs	$43,700	
Annual Maintenance Costs	$5,200	$100 hosting fee / month * 12 months + $4000 annual maintenance staff budget

Projected Timeline

The following table presents a projected timeline of the deliverables and milestones, with their start and end dates, dependencies (prerequisites), and resources they are assigned. The employees will work typical 8-hour long workdays Monday-Friday throughout seven weeks total.

#	Prerequisites	Milestones	Resources	Days	Start date	End date
1	-	Meet with stakeholders and obtain project approval	Project Manager, AI Software Engineer, Full-Stack Software Engineer	3	06/03/24	06/05/24
2	-	Define technical constraints	AI Software Engineer, Full-Stack Software Engineer	1	06/06/24	06/07/24
3		Purchase equipment	Project Manager	1	06/10/24	06/11/24
4	1, 2	Plan algorithm, data, and app access specifications	AI Software Engineer, Full-Stack Software Engineer	2	06/12/24	06/14/24
5	3	Create a software design	AI Software Engineer, Full-Stack Software Engineer	2	06/17/24	06/19/24
6	3	Process data	Full-Stack Software Engineer	1	06/20/24	06/21/24
7	4	Develop a machine learning model	AI Software Engineer	4	06/24/24	06/28/24
8	6	Add a form and data visualization tools	AI Software Engineer	2	07/01/24	07/03/24
9	6	Test the machine learning model	AI Software Engineer	1	07/04/24	07/05/24
10	8	Test form functionality	Full-Stack Software Engineer	1	07/08/24	07/09/24
11	9	Test application as a whole	Full-Stack Software Engineer	2	07/10/24	07/12/24
12	10	Deploy the application with documentation	Full-Stack Software Engineer	2	07/15/24	07/17/24
13	11	Continue maintenance	AI Software Engineer, Full-Stack Software Engineer	-	07/28/24	Onwards

Part C: Application

This is the link to the application: https://mybinder.org/v2/gh/ari-abr/capstone.git/HEAD . Please note that it can take over a few minutes for it to load. The main project is stored inside the "Capstone.ipynb" file.

Part D: Post-implementation Report

Business Vision and a Solution Summary

The problem was that as cardiovascular health services grow in demand, medical practitioners need better technology to assist them in making accurate and well-informed decisions, supporting the patient as well. The solution to this problem was creating an application utilizing machine learning to predict if a patient is likely to suffer heart failure using the patient's medical data from a form. This application has the potential to benefit medical practitioners by aiding in the overall treatment efficiency and accuracy.

The solution was made using Python 3 programming language and libraries including Pandas and Seaborn for data visualization, Sklearn for the machine learning model, and Ipywidgets for the user interface and the form. It was developed using Jupyter Notebook and hosted using the MyBinder online environment. The application itself consisted of the main Jupyter Notebook Python (.ipynb) file and a .csv data file inside one directory.

Data Summary and Source

The raw data was collected from an online free source, Kaggle. During the design stage, the project laid out data requirements and constraints, including containing numerical data and having multiple columns for independent variables and one column for dependent variables (Chicco & Jurman, 2020). Then, the dataset was downloaded from the source as a .csv file and checked to ensure that all of its fields were filled out and that it did not contain any missing data. During the implementation, the data was loaded into the main application file using Python and data frames. For deployment, the data file was stored and deployed in the same directory as the main application file.

Raw and Cleaned Datasets

This project did not involve cleaning or additional processing of the data since the data that was downloaded was already ready to be used. The screenshot below shows the raw data that was downloaded from the source. All of the fields are represented in numerical format. The columns include: age, anaemia, creatinine_phosphokinase, diabetes, ejection_fraction, high_blood_pressure, platelets, serum_creatinine, serum_sodium, sex, smoking, time, and DEATH_EVENT.

age	anaemia	creatinine_phosphokinase	diabetes	ejection_fraction	high_blood_pressure	platelets	serum_creatinine	serum_sodium	sex	smoking	time	DEATH_EVENT
75	0	582	0	20	1	265000	1.9	130	1	0	4	1
55	0	7861	0	38	0	263358.03	1.1	136	1	0	6	1
65	0	146	0	20	0	162000	1.3	129	1	1	7	1
50	1	111	0	20	0	210000	1.9	137	1	0	7	1
65	1	160	1	20	0	327000	2.7	116	0	0	8	1
90	1	47	0	40	1	204000	2.1	132	1	1	8	1
75	1	246	0	15	0	127000	1.2	137	1	0	10	1
60	1	315	1	60	0	454000	1.1	131	1	1	10	1
65	0	157	0	65	0	263358.03	1.5	138	0	0	10	1
80	1	123	0	35	1	388000	9.4	133	1	1	10	1
75	1	81	0	38	1	368000	4	131	1	1	10	1
62	0	231	0	25	1	253000	0.9	140	1	1	10	1
45	1	981	0	30	0	136000	1.1	137	1	0	11	1
50	1	168	0	38	1	276000	1.1	137	1	0	11	1
49	1	80	0	30	1	427000	1	138	0	0	12	0
82	1	379	0	50	0	47000	1.3	136	1	0	13	1

Code for Data Analysis and Constructing A Predictive Product

This application was coded in Python. The code for creating a machine learning model and model accuracy is shown in the screenshot below. First, the data was loaded as a data frame and split into independent and dependent variables. Then, a logistic model was created using the Sklearn library and conducted the testing of the data subsets using the train_test_split() function. Then the accuracy_score() function was used to measure the accuracy of the prediction.

```
# Separate the data into independent and dependent variables
y = df.values[:,12]
x = df.values[:,0:12]

# Train model with supervised learning (logistic regression)
mylog_model = linear_model.LogisticRegression(max_iter=10000)
x_train, x_test, y_train, y_test = model_selection.train_test_split(x,y,test_size=0.2)
mylog_model.fit(x_train, y_train)

# Make a prediction and measure its accuracy
y_pred = mylog_model.predict(x_test)
print("The accuracy of this model is: " + str(sklearn.metrics.accuracy_score(y_test,y_pred)))
```

The code for the descriptive methods is included below. The confusion matrix plot was made using the Sklearn library and the model made earlier. The logistic regression graph was made using the Seaborn library and data frame. The histograms were made using the Pandas library.

```
# Confusion matrix
cm = confusion_matrix(y_test, y_pred)
disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
```

```
# Logistic Regression
sns.regplot(x='age', y='DEATH_EVENT', data=df, logistic=True)
```

```
# Histograms
df.hist(figsize=(20, 10))
```

The code for the predictive method used the machine learning model made earlier to make a prediction based on the form input. The form was made using the Ipywidgets library, providing a simple user interface with widgets that the user can use to select a choice or enter a number. Each of these widgets has a value field that holds the default or an updated entered value.

At the end of the form, the submission button has a function connected to it so that when it gets clicked, it calls the make_prediction() function. The key part of the make_prediction() function is shown in the screenshot below. First, the function makes an array of each of the widget values. Then the logistic regression model makes a prediction based on that value array. Last, the prediction result is displayed inside the output window to the user.

```
# Predict the value using logistic regression model and form inputs
values = [ageWidget.value, anaemiaInput, creatinineWidget.value, diabetesInput, ejectionWidget.value,
          pressureInput, plateletsWidget.value, serumcreatinineWidget.value, serumsodiumWidget.value,
          genderInput, smokingInput, 0]
print(values)
prediction_num = mylog_model.predict([values])

# Print the results to the user
print('Prediction result value: ' + str(prediction_num))
if prediction_num == 1: prediction_text = 'likely to have a heart failure.'
else: prediction_text = 'unlikely to have a heart failure.'
print('The patient will be: ' + str(prediction_text))
```

Machine Learning

The method for machine learning is a machine-supervised learning algorithm with logistic regression. The method was developed by using Python with a machine-learning library called Sklearn. A supervised learning method was chosen because it is easier to implement and the logistic regression was chosen because it is commonly used for dependent variables that fall into one of two categories, which works in this scenario since the dependent variable in this project is whether or not a heart failure will occur.

Validation and Assessment of the Hypotheses

The hypothesis was that if a heart failure detection model is implemented using a machine learning logistic regression model, it will have an average accuracy score of at least 75% when presented with sample cases. The machine learning algorithm tested this hypothesis by using a Sklearn library function to calculate the accuracy: sklearn.metrics.accuracy_score(y_test,y_pred).

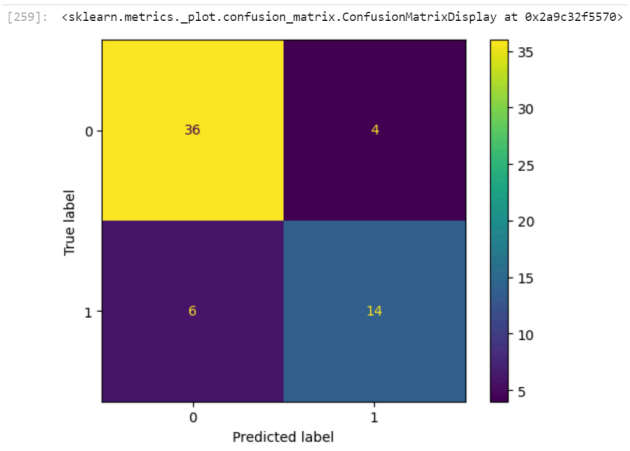
This accuracy number did vary from time to time, so a sample of ten consecutive trials and their accuracy score was recorded in the table below. The average accuracy score was about 0.81 which meets the goal of having accuracy above 75% (above 0.75). This shows that the hypothesis has been accepted.

Trial	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
Result	0.85	0.77	0.70	0.83	0.78	0.82	0.85	0.83	0.88	0.80

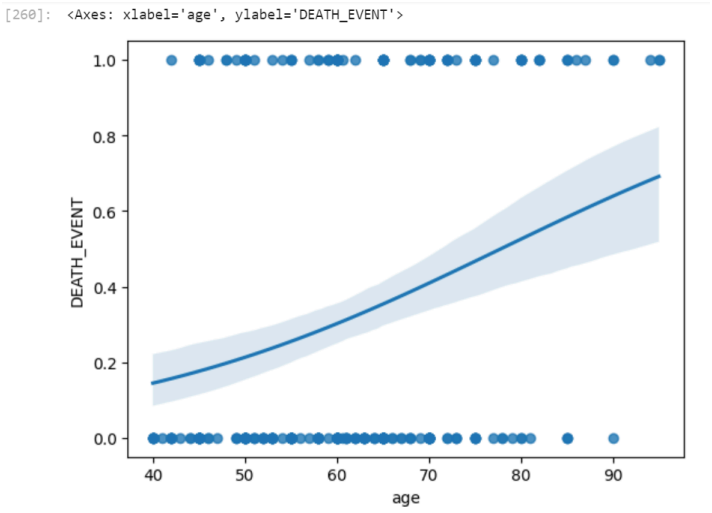
Visualizations

All of the visualization tools are located within the “Capstone.ipynb” file.

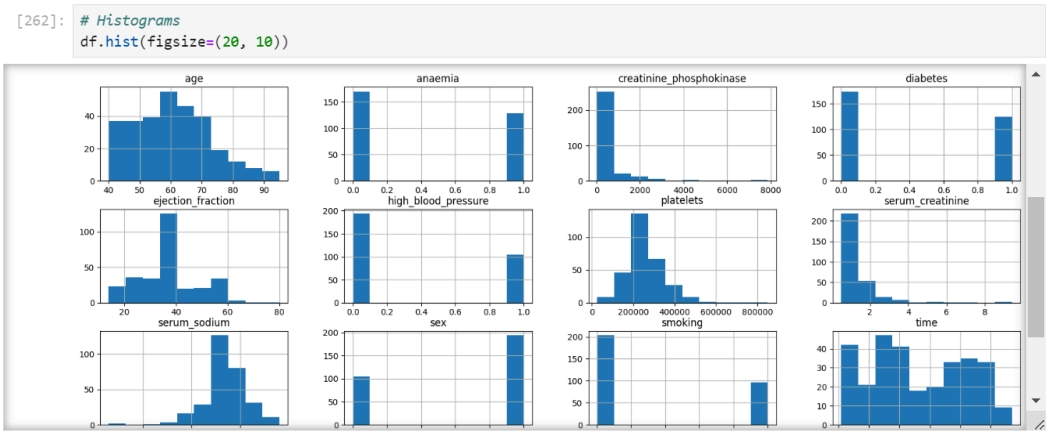
The first visualization tool was a confusion matrix, depicting the performance of the logistic regression model. The matrix below shows that in most cases the true label and predicted label align, demonstrating that the logistic regression model has high accuracy, aiding in data analysis.



The second visualization tool was a logistic regression curve graph, showing the relationship between age and whether heart failure has occurred. This helps visualize the data phenomenon of how heart failure can correlate with other medical factors.



The third visualization tool was histograms. These graphs allow data exploration and summary by showing the distribution of each of the medical factor's values. This helps the user have a better understanding of the data that was used for training the model.



Assessment of the Product's Accuracy

This product's accuracy was sufficient for this project, however, it has areas for improvement to be used in a real-life medical setting. The product does meet its goal of having an average accuracy of about 81%, indicating that in most cases it does make the correct prediction based on similar data it has been used for training, which satisfies the project's goal and hypothesis.

However, in a real clinical environment, this number may not be sufficient since the medical field aims to have accuracy as close to 100% as possible as people's lives can be at stake. To improve the product's accuracy, more additional medical factors can be included such as data covering a wider range of medical factors and demographics, as well as further adjusting and refining the machine learning algorithm.

Results from the Data Product Testing, Revisions, and Optimization

The first round of testing was done during the implementation phase. First, the data was checked to verify that does not have any formatting or spelling variations or missing fields. Then the different algorithms besides logistic, were tested such as linear regression and Support Vector Machine. However, they tended to have lower accuracy scores than logistic, so the logistic regression was chosen. Additionally, the model's testing size was decreased to 0.2 for optimization. Increasing up to over 0.5 or decreasing it down to 0.1 both showed to decrease the average accuracy score from about 0.8 to about 0.6-0.7.

```
# Train model with supervised learning (Logistic regression)
mylog_model = linear_model.LogisticRegression(max_iter=10000)
x_train, x_test, y_train, y_test = model_selection.train_test_split(x,y,test_size=0.9)
mylog_model.fit(x_train, y_train)

# Make a predicton and measure its accuracy
y_pred = mylog_model.predict(x_test)
print("The accuracy of this model is: " + str(sklearn.metrics.accuracy_score(y_test,y_pred)))
```

```
The accuracy of this model is: 0.7074074074074074
```

After the machine learning model was optimized, it was tested to obtain an average accuracy score of at least 75% to verify that the model met the goal.

Then after the form and data visualization tools were built, they were tested to ensure that they were functional and displayed appropriately. The form was tested by entering different inputs and clicking the submission button to see that the result would be displayed. These user interface features were revised for better clarity and usability.

After implementation, the application was tested as a whole to ensure that no parts of the code resulted in warnings or errors.

Last, after the deployment, the application was tested once again to verify that it was displayed appropriately and was functional using the MyBinder link provided.

Source Code and Executable files

The source code can be accessed following the link:

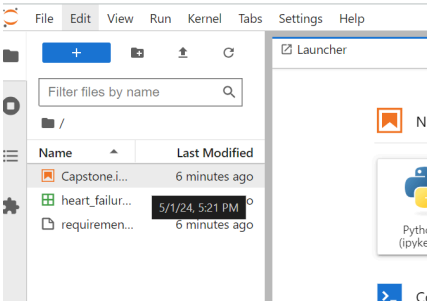
https://mybinder.org/v2/gh/ari-abr/capstone.git/HEAD .

The executable file is “Captstone.ipynb”.

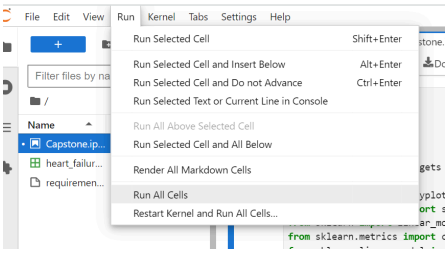
User Guide

Instructions:

1) Click on the link to launch the application: https://mybinder.org/v2/gh/ari-abr/capstone.git/HEAD. Please note that it may take a few minutes for the application to load.
2) In the file directory on the left side, click on the “Captstone.ipynb” file.

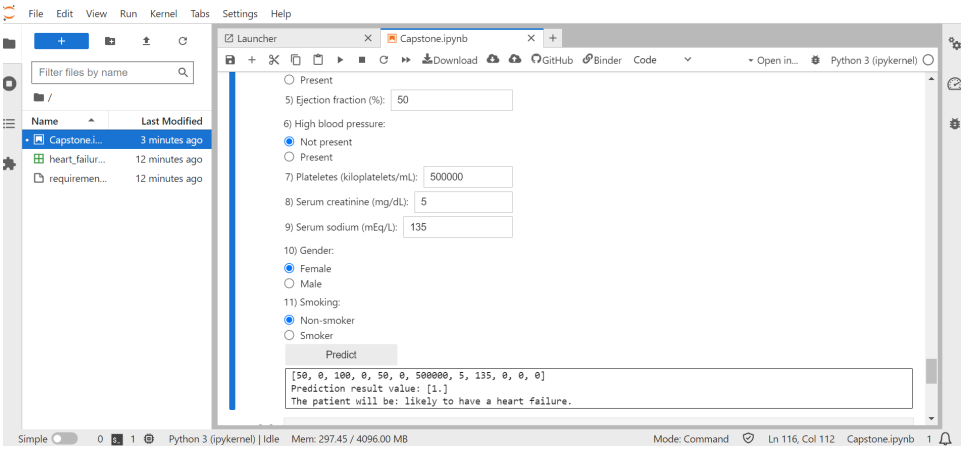
a) 

3) In the top menu, click on “Run”, then “Run All Cells”. You may need to wait about a minute for all visual parts to load as they may not load instantly. If any of the cells still do not load, run each cell one by one starting from the top.

a) 

4) Scroll down to view the visual graphics.

5) To use the interactive form, scroll down to the form under the “Heart Failure Prediction Form” heading. Use the widgets to enter form information and click on the “Predict” button to view the result.

a) 

6) To rerun the application, click on “Run”, and then “Run All Cells” all again.

The following is an example of a client using the application:

1) A medical practitioner accesses this application prior to the patient's examination.

2) During the patient examination, the practitioner gathers medical information needed such as blood test levels.

3) The practitioner opens and accesses the app and scrolls down to the form.

4) The practitioner enters the patient’s data and clicks on the “Predict” button.

5) The patient views the prediction and takes the appropriate actions for the medical care needed based on it.

Reference Page

Chicco, D., & Jurman, G. (2020). Machine Learning Can Predict Survival of Patients with Heart Failure from Serum Creatinine and Ejection Fraction Alone. BMC Medical Informatics and Decision Making, 20(1). https://doi.org/10.1186/s12911-020-1023-5